

STAT 9120: Theory of Linear Models

Homework 1: Linear Algebra Foundations

Week 1

Instructions. Show all work and justify your answers. Unless a problem specifically asks for numerical computation, answers consisting only of a final numerical result will receive limited credit. You may use standard matrix arithmetic software to check your calculations, but your submitted work should demonstrate the relevant linear algebra directly.

Problem 1. Matrix Arithmetic and Linear Combinations

10 points

Let

$$A = \begin{pmatrix} 1 & 2 \\ 0 & 1 \\ 2 & -1 \end{pmatrix}, \quad x = \begin{pmatrix} 3 \\ -2 \end{pmatrix}.$$

- (a) Compute Ax . [4 points]
- (b) Let a_1 and a_2 denote the columns of A . Express Ax as a linear combination of a_1 and a_2 . [3 points]
- (c) Explain briefly why $Ax \in \mathcal{C}(A)$ for every $x \in \mathbb{R}^2$. [3 points]

Problem 2. Linear Independence and Rank

12 points

Consider the vectors

$$x_1 = \begin{pmatrix} 1 \\ 0 \\ 1 \end{pmatrix}, \quad x_2 = \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix}, \quad x_3 = \begin{pmatrix} 2 \\ -1 \\ 1 \end{pmatrix}.$$

- (a) Determine whether x_1, x_2, x_3 are linearly independent. Justify your answer. [6 points]
- (b) Let

$$X = \begin{pmatrix} x_1 & x_2 & x_3 \end{pmatrix}.$$

- Determine $\text{rank}(X)$. [3 points]
- (c) Give a basis for $\mathcal{C}(X)$. [3 points]

Problem 3. Null Space and Rank–Nullity

12 points

Let

$$A = \begin{pmatrix} 1 & 2 & -1 \\ 2 & 4 & -2 \end{pmatrix}.$$

- (a) Find $\mathcal{N}(A)$ and give a basis for it. [5 points]
- (b) Determine $\text{rank}(A)$. [3 points]
- (c) Determine $\dim \mathcal{N}(A)$. [2 points]
- (d) Verify explicitly that the rank–nullity theorem holds for A . [2 points]

Problem 4. $X^T X$, Rank, and Positive Definiteness**12 points**Let $X \in \mathbb{R}^{n \times p}$.

- (a) Prove that

$$\boxed{\mathcal{N}(X^T X) = \mathcal{N}(X)}.$$

Your argument should use the identity

$$a^T X^T X a = \|Xa\|^2.$$

[6 points]

- (b) Use part (a) to show that

$$\text{rank}(X^T X) = \text{rank}(X).$$

[3 points]

- (c) Show that if
- X
- has full column rank, then
- $X^T X$
- is positive definite. [3 points]

Problem 5. Column Spaces and Orthogonal Complements**10 points**

Let

$$X = \begin{pmatrix} 1 & 0 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

- (a) Find a basis for
- $\mathcal{C}(X)$
- . [3 points]

- (b) Find all vectors
- $z \in \mathbb{R}^3$
- satisfying

$$X^T z = 0.$$

Give a basis for $\mathcal{N}(X^T)$. [4 points]

- (c) Verify directly in this example that

$$\mathcal{C}(X)^\perp = \mathcal{N}(X^T).$$

[3 points]

Problem 6. Projection onto a One-Dimensional Subspace**10 points**

Let

$$u = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$$

and let $S = \text{span}(u)$. The orthogonal projection of $y \in \mathbb{R}^2$ onto S is

$$P_S y = \frac{uu^T}{u^T u} y.$$

(a) Find the matrix P representing the projection onto S . [4 points]

(b) Verify that $P^T = P$. [2 points]

(c) Verify that $P^2 = P$. [2 points]

(d) For

$$y = \begin{pmatrix} 3 \\ 1 \end{pmatrix},$$

compute Py and $y - Py$, and verify that they are orthogonal. [2 points]

Problem 7. The Least-Squares Projection Matrix

14 points

Suppose $X \in \mathbb{R}^{n \times p}$ has full column rank, and define

$$H = X(X^T X)^{-1} X^T.$$

(a) Show that $H^T = H$. [3 points]

(b) Show that $H^2 = H$. [4 points]

(c) Show that $HX = X$. [3 points]

(d) Show that $X^T(I - H) = 0$. [2 points]

(e) Explain geometrically what parts (c) and (d) tell us about the spaces onto which H and $I - H$ project. [2 points]

Problem 8. Least Squares as Orthogonal Projection

10 points

Consider the linear model

$$Y = X\beta + \varepsilon,$$

where X has full column rank. Define

$$\hat{\beta} = (X^T X)^{-1} X^T Y, \quad \hat{Y} = X\hat{\beta}, \quad e = Y - \hat{Y}.$$

(a) Show that $X^T e = 0$. [3 points]

(b) Explain why part (a) implies $e \in \mathcal{C}(X)^\perp$. [2 points]

(c) Explain why $\hat{Y} \in \mathcal{C}(X)$. [2 points]

(d) Use the preceding results to establish the decomposition

$$\|Y\|^2 = \|\hat{Y}\|^2 + \|e\|^2.$$

State the geometric principle underlying this equality. [3 points]

Problem 9. Eigenvalues and Projection Matrices**10 points**

Let P be an $n \times n$ idempotent matrix, so that

$$P^2 = P.$$

(a) Show that every eigenvalue of P must be either 0 or 1. [5 points]

(b) Suppose in addition that P is symmetric. Explain why

$$\text{tr}(P) = \text{rank}(P).$$

[3 points]

(c) If $H = X(X^T X)^{-1} X^T$ and X has rank p , determine $\text{tr}(H)$ and $\text{tr}(I - H)$. [2 points]

Problem 10. Rank Deficiency and Identifiability**10 points**

Consider the design matrix

$$X = \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & 3 \end{pmatrix}.$$

(a) Show that the columns of X are linearly dependent. [2 points]

(b) Find a nonzero vector a such that $Xa = 0$. [2 points]

(c) Show that for every parameter vector $\beta \in \mathbb{R}^3$ and every scalar c ,

$$X(\beta + ca) = X\beta.$$

[3 points]

(d) Explain what part (c) means for the identifiability of β . Why can the mean vector $X\beta$ nevertheless still be uniquely defined? [3 points]

Total: 100 points**Concepts to Review**

Before completing the assignment, you should be comfortable with the following ideas:

$$\mathcal{C}(X), \quad \mathcal{N}(X), \quad \text{rank}(X),$$

linear independence, basis and dimension,

$$\mathcal{C}(X)^\perp = \mathcal{N}(X^T),$$

orthogonal projection,

$$P^2 = P, \quad P^T = P,$$

the least-squares projection matrix

$$H = X(X^T X)^{-1} X^T,$$

and the relationship between rank deficiency and parameter identifiability.